

Large-scale text indexing and search with the Jaccard index and MinHash

Kristoffer Sahlin

Department of Mathematics, Science for Life Laboratory,
Stockholm University

Why are text searches needed?

- Information search
- Document Clustering
- Plagiarism detection
- Bioinformatics
- ...

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Documents / Webpages

- 0** Climate change is one of the most pressing challenges facing modern society, affecting ecosystems, economies, and communities across the globe...
- 1** Climate change is among the most urgent challenges confronting modern society, influencing ecosystems, economies, and communities worldwide. Recently, scientific studies have...
- 2**
- :**
- :**
- n** Climate change represents a major global challenge in the twenty-first century, shaping ecosystems, economies, and societies around the world. In recent years, scientific studies...

Why are text searches needed?

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Bacteria ID

```
0 CACGACTCTGGTACCTAGACTCGACGATCGTACTGCTGCTGTCAGATCGATCGAGTCGACCCACGT....
1 CACGACTCTGGTACCTAGACTTCGACGTCGAGTCGTACGCTGTCAGATCGAGCCTCGAGTCGACCGA....
2 TTACGACTCTGGTACCTAGATTTCGACGATCGTACCCTGCTGCTGTCAGATCGCCTCGAGTCGTACTG....
:
:
n TGTTTTTACGACTCTGGTACCTAGACTCGACGATCGTACCCTGCTGCTGTCAGATCGAGCCTCACTA....
```

- **Bacteria length:** ~2–6 million letters
- **n:** in the millions
- Terabytes of text

How to compare two DNA sequences?

CACTCCTAC CACTCTCTA

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| | | | | x x x x
CACTCTCTA

Hamming distance: 4

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CACTCTCTA

Hamming distance: 4

CACTC-CTAC
| | | | | x | | | x
CACTCTCTA-

Edit distance: 2
(Levenshtein distance)

Problem: unordered local similarities

Bacteria ID

0 CACGACTCTGGTACCTAGACTCGACGTCGGTACTGCTGCTGTCAGATCGATCGAGTCGACCCACGT....
1 CACGACTCTGGTACCTAGACTTCGACGTCGAGTCGTACGCTGTCAGATCGAGCCTCGAGTCGACCGA....
2 TACCCTGCTGCTGTCAGATCGCCTCGAGTCGTACTG TTACGACTCTGGTACCTAGATTTCGACGATCG....
:
:
n TTACGACTCTGGTACCTAGATTTCGACGATCG TACCCTGCTGCTGTCAGATCGCCTCGAGTCGTACTG...

Solution: Compare substrings (*k*-mers)

Bacteria ID

0 CACGACTCTGGTACCTAGACTCGACGTCGGTACTGCTGCTGTCAGATCGATCGAGTCGACCCACGT....
1 CACGACTCTGGTACCTAGACTTCGACGTCGAGTCGTACGCTGTCAGATCGAGCCTCGAGTCGACCGA....
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:
:
n TTACGACTCTGGTACCTAGATTTCGACGATCG TACCCTGCTGCTGTCAGATCGCCTCGAGTCGTACTG...

TTACGACTC
TACGACTCT
ACGACTCTG
CGACTCTGG

...

CGTACCCTG
GTACCCTGC
TACCCTGCT
ACCCTGCTG



k-mers (here $k = 9$)

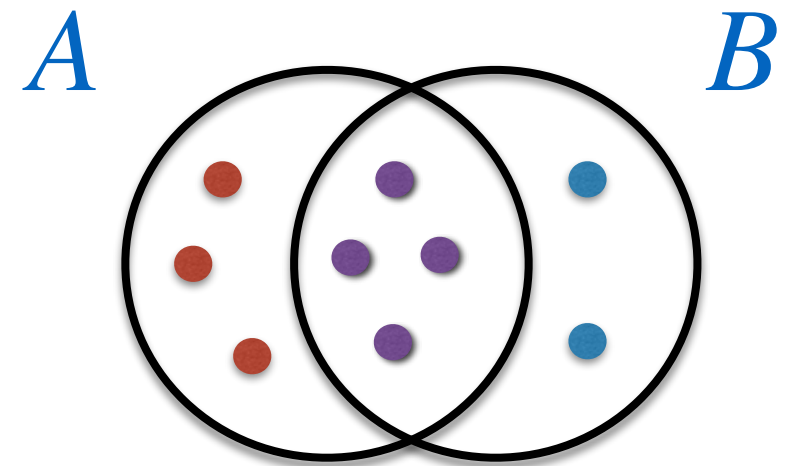
The *Jaccard index*

A and B sets of items

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

$$0 \leq J(A, B) \leq 1$$

An illustration



$$J(A, B) = \frac{|\text{Intersection}|}{|\text{Union}|} = \frac{4}{9}$$

Example: Jaccard index for DNA sequences

Hamming distance

Edit distance

Previously

CACTCCTAC

|||||xxxxx

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CACTC-CTAC

|||||x||||x

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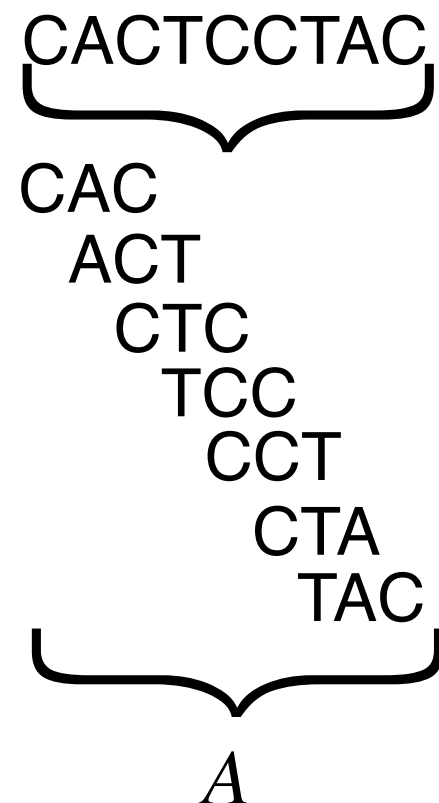
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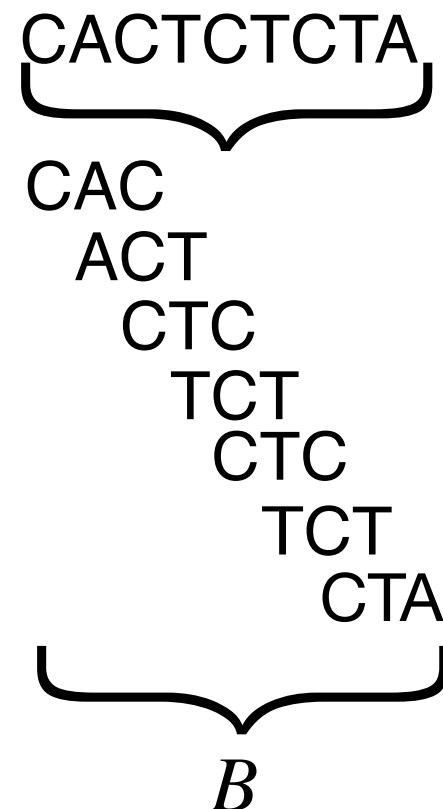
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Jaccard index ($k = 3$)



k-mers



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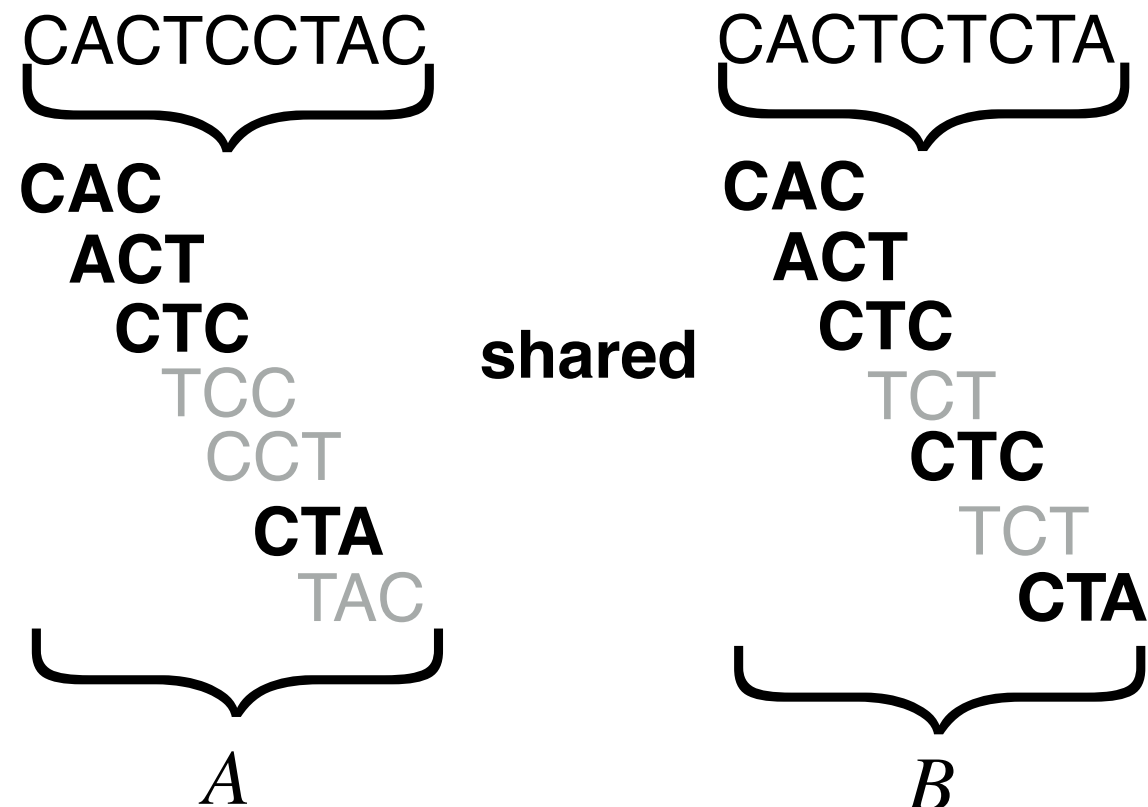
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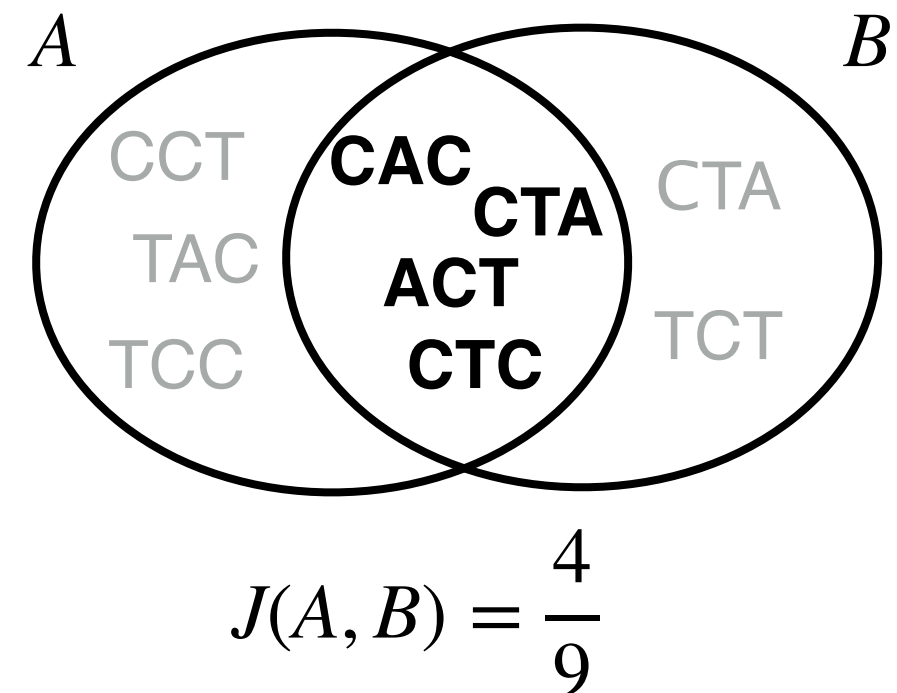
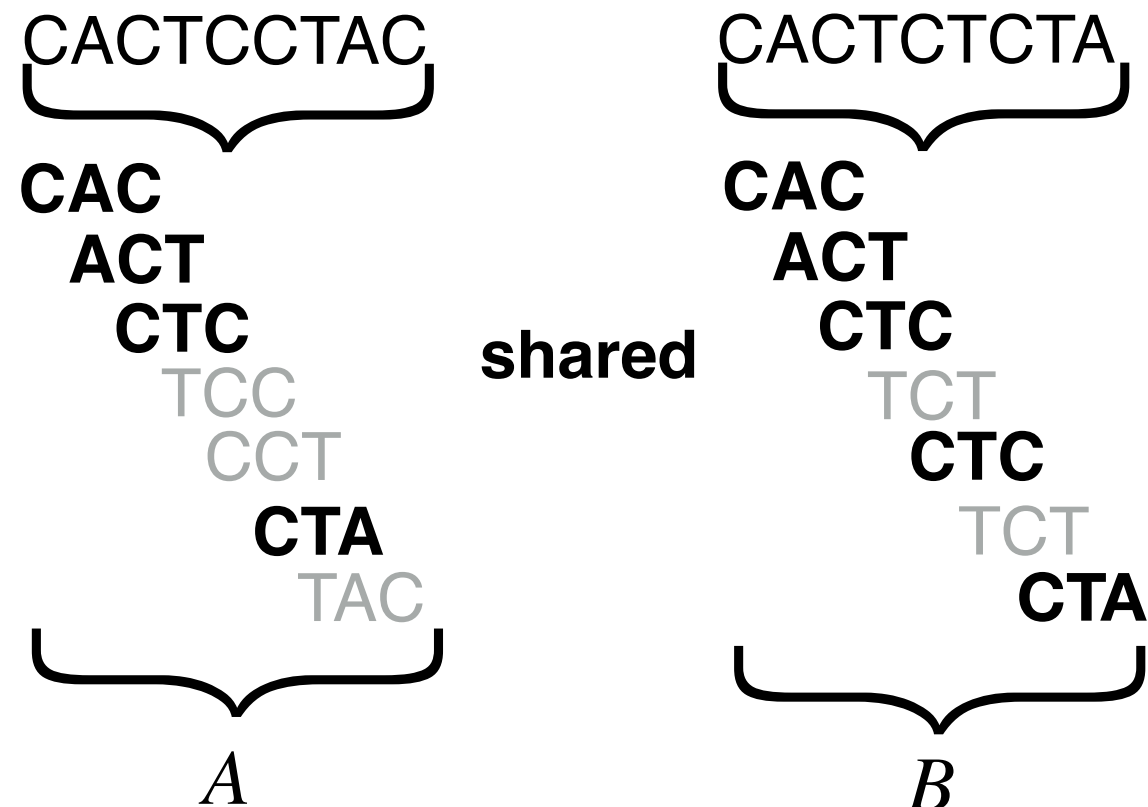
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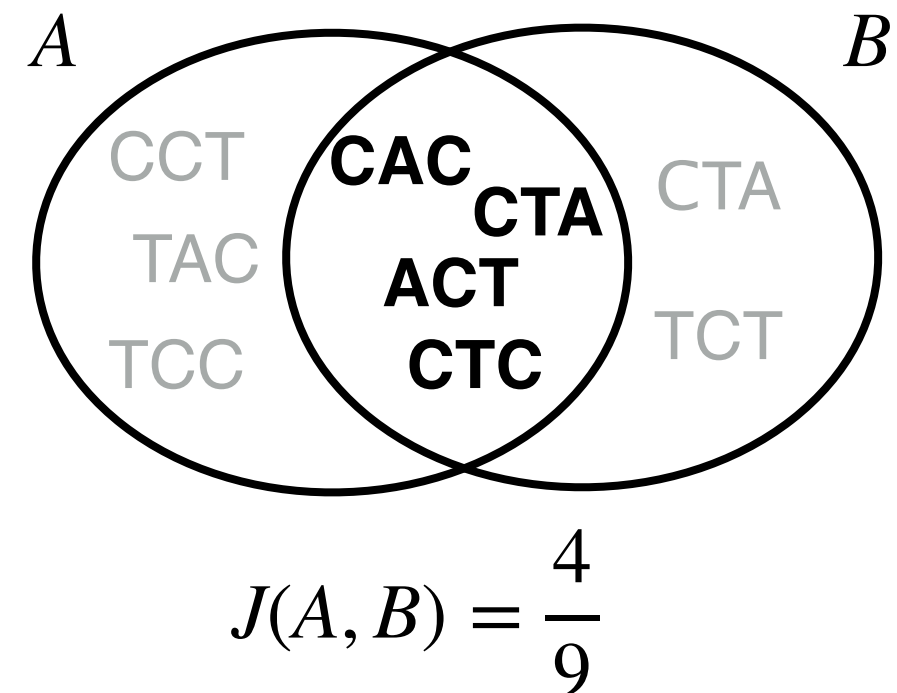
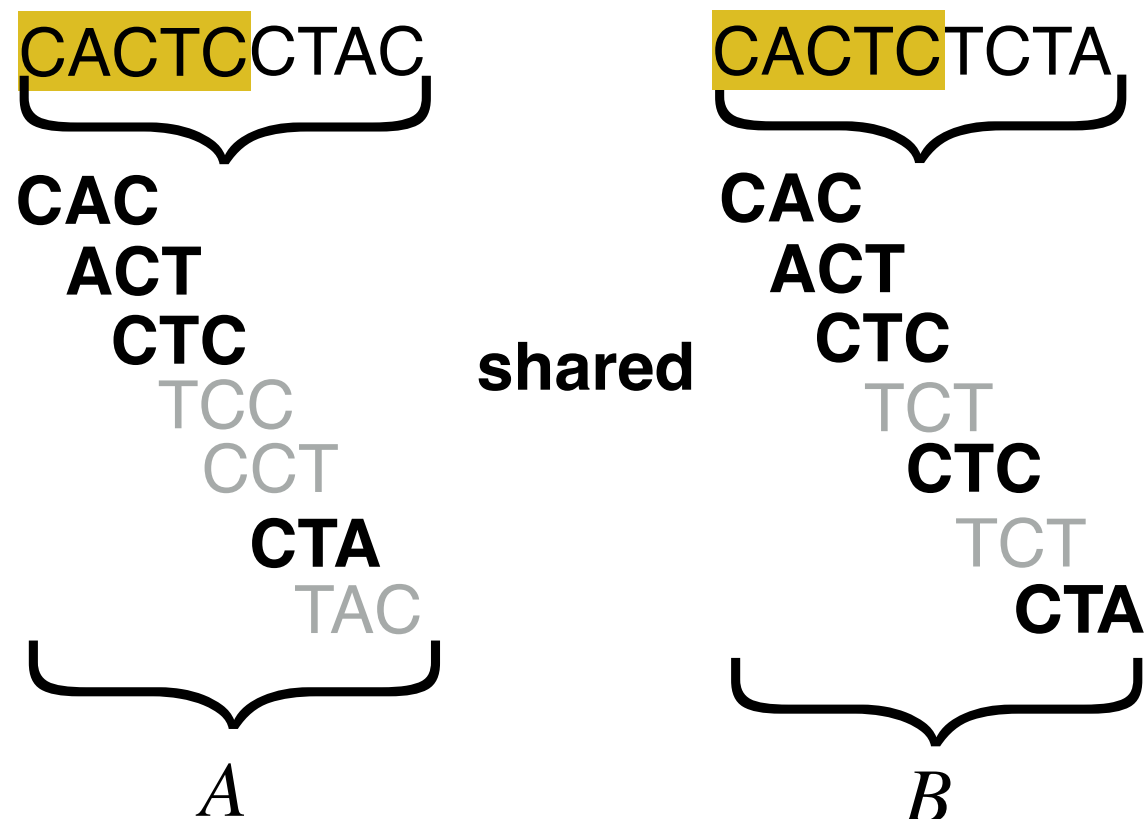
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Jaccard index ($k = 3$)


Problem solved: Local similarity detected!

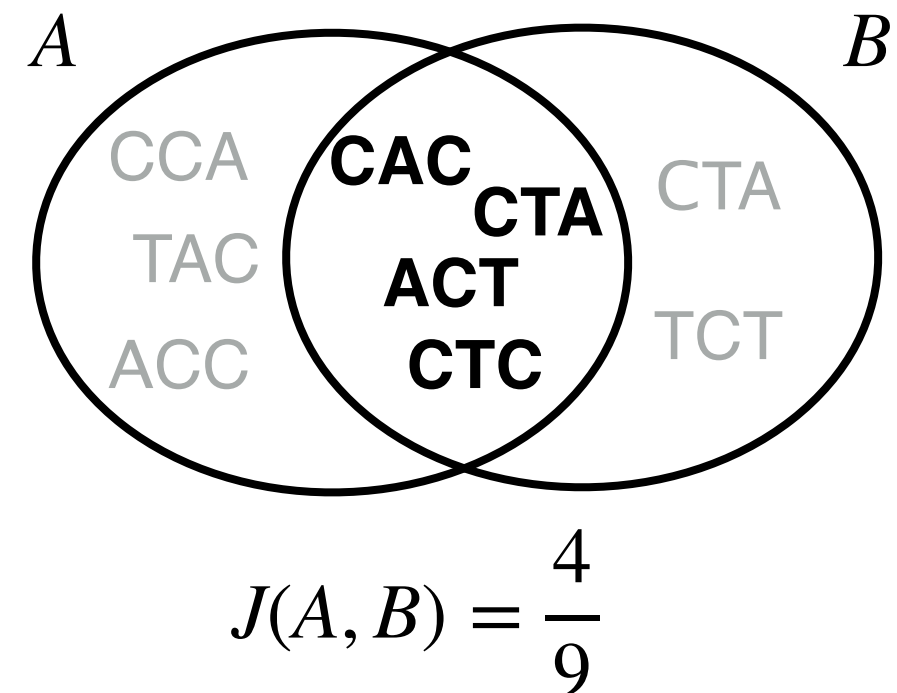
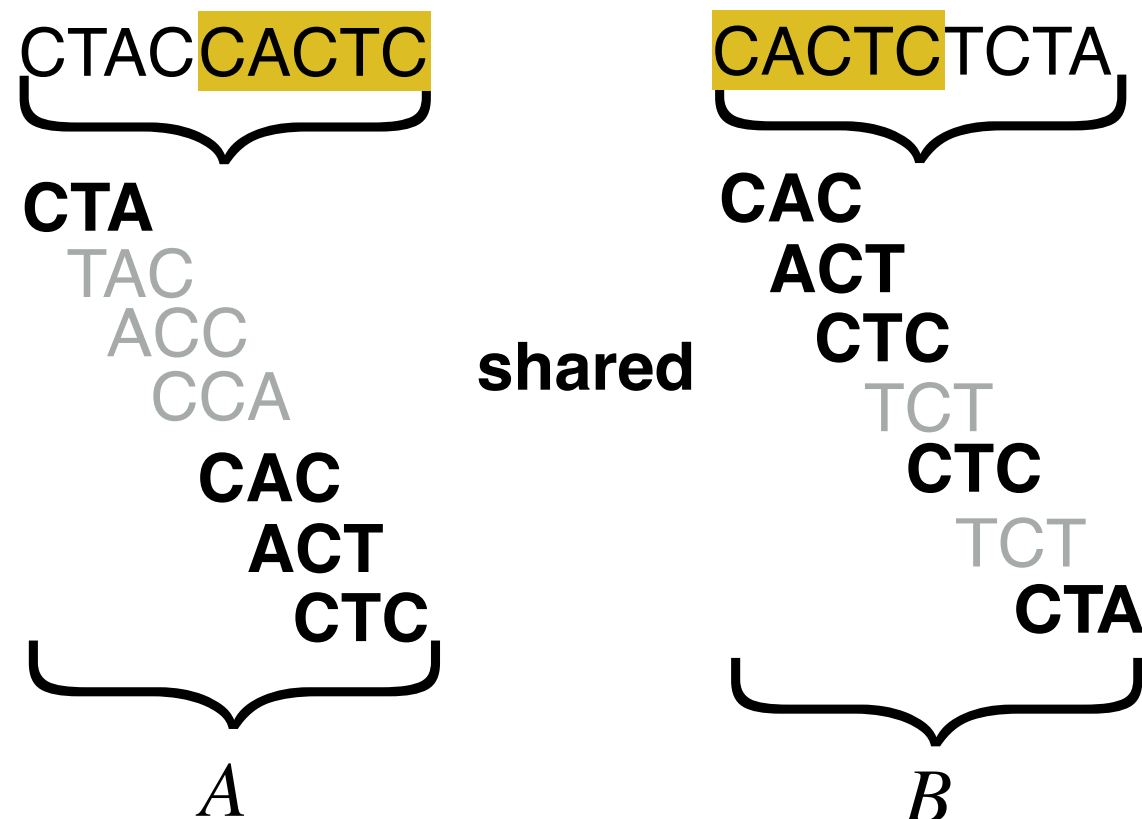
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Jaccard index ($k = 3$)


Jaccard index problem: Time and memory intensive

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ACT
CTC
TCC
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k-mer redundancy!

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k-mer redundancy!

- Idea: "sample" only some *k*-mers!
- Criterion: Same input gives same output (deterministic)
- Called *sketching* (or *fingerprinting*)

MinHash¹ sketching: deterministic sampling

Step 1: Map each k -mer to a random integer (hash function)

Step 2: Keep the minimum integer (this is a **MinHash** value)

Step 3: Repeat with m independent random mappings

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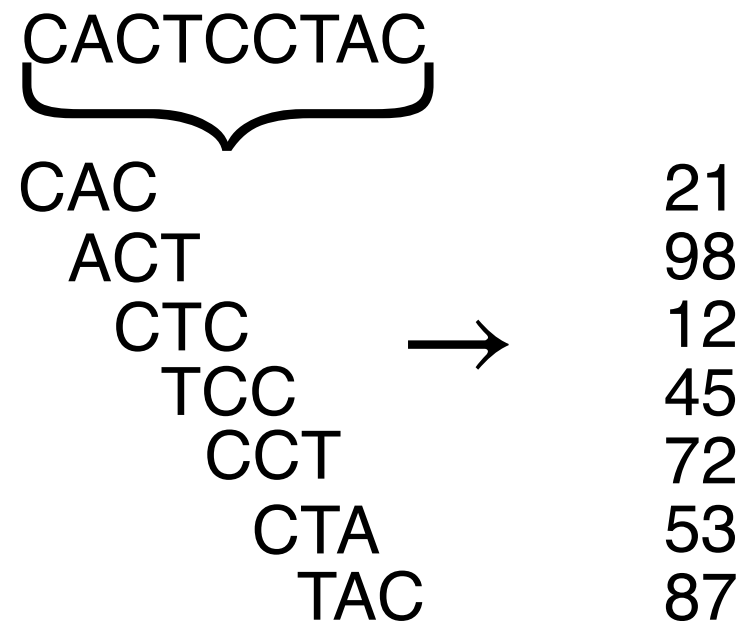
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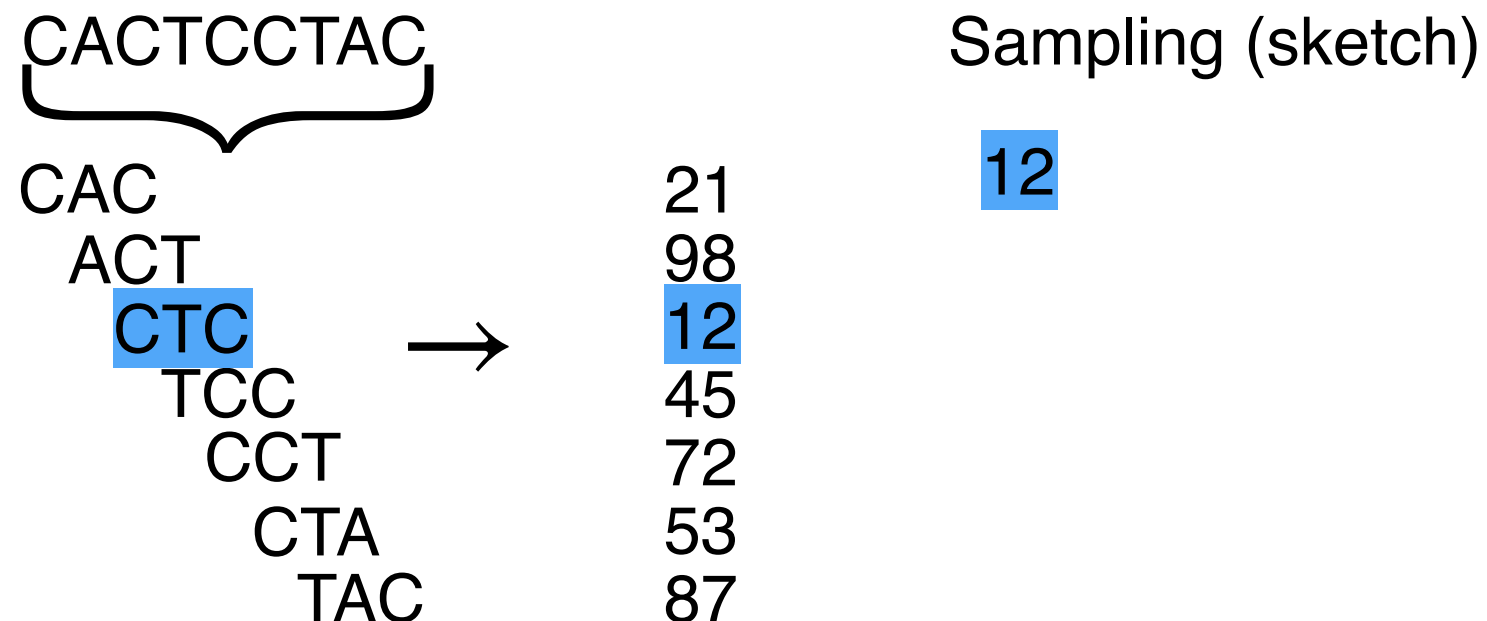
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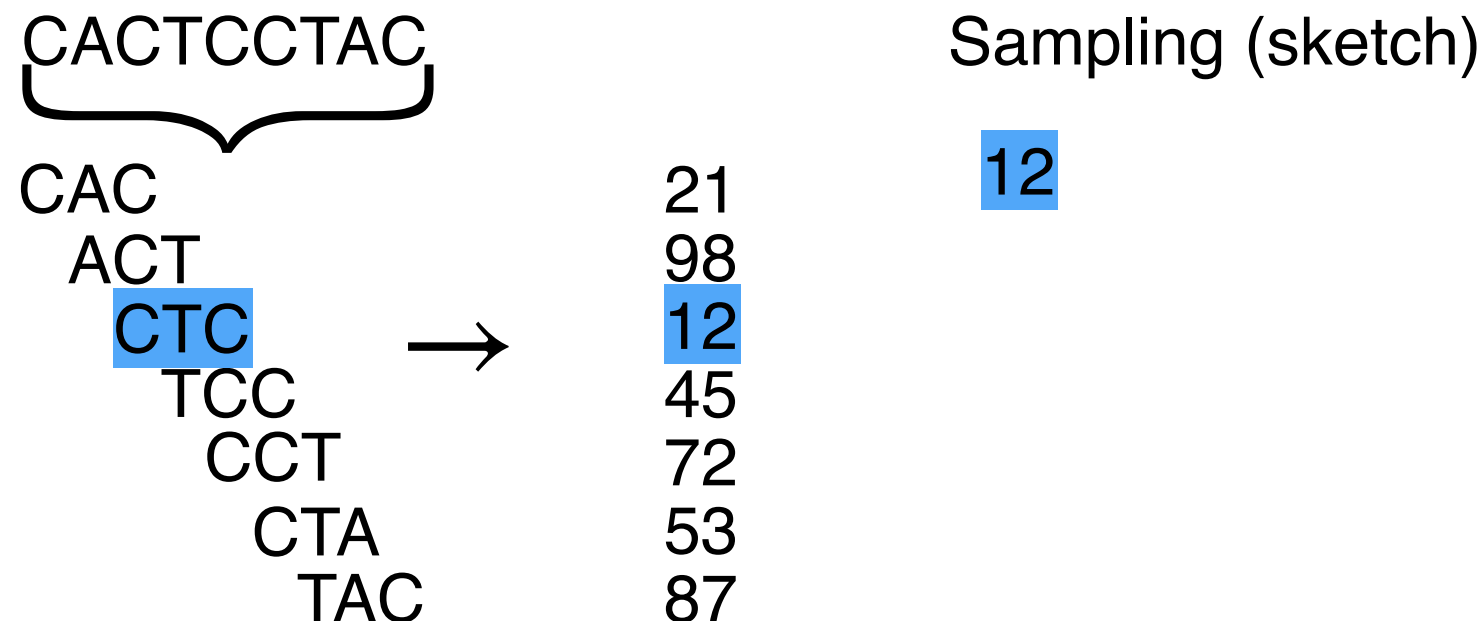
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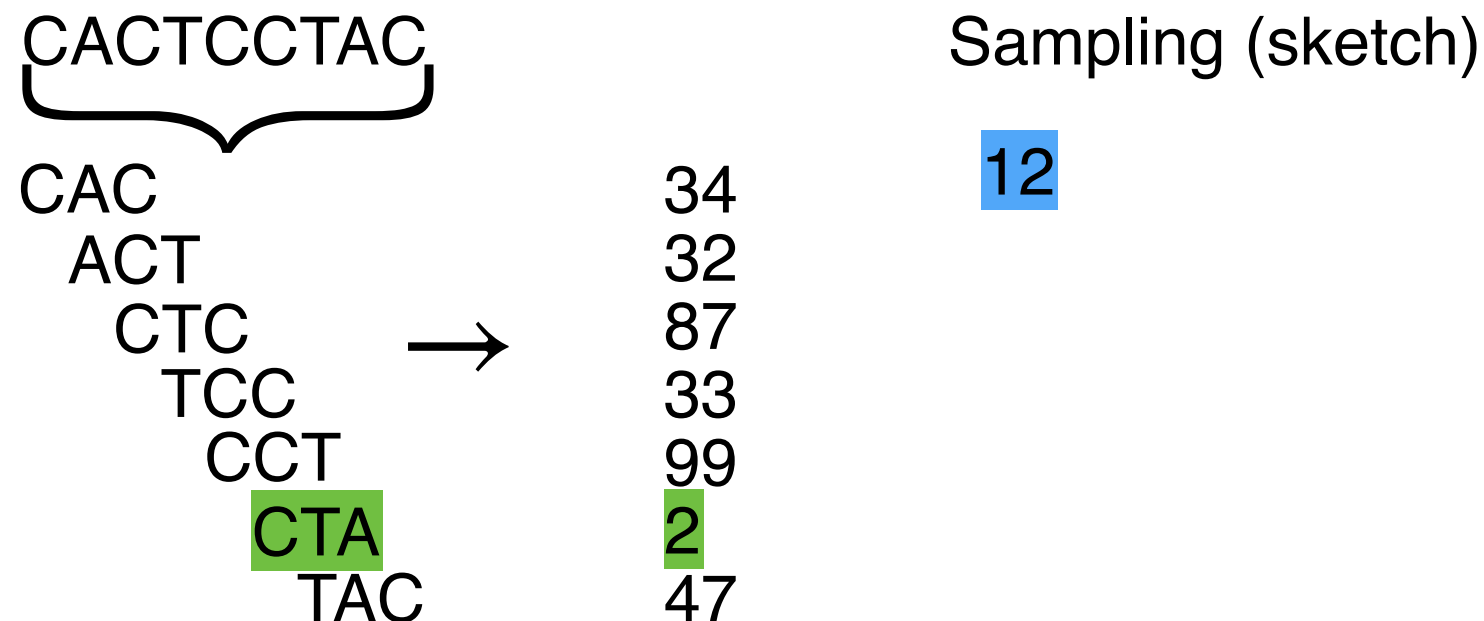
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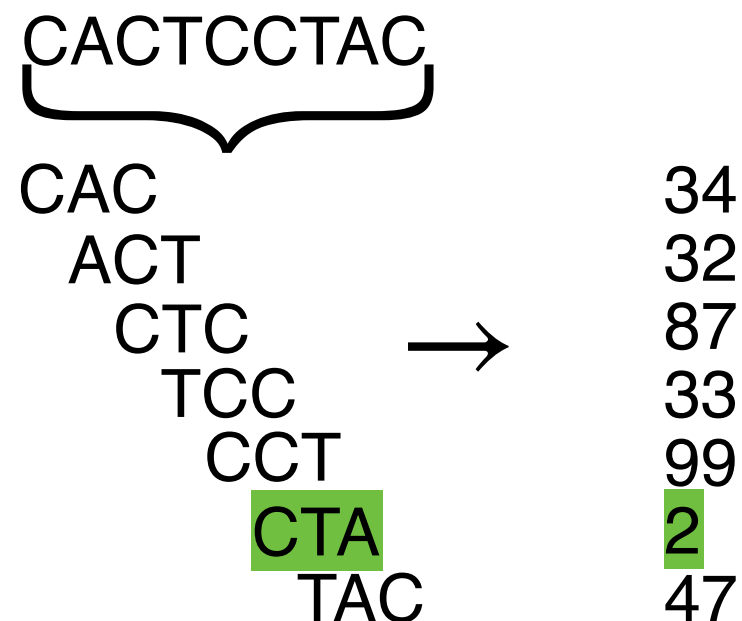
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Sampling (sketch)

12, 2

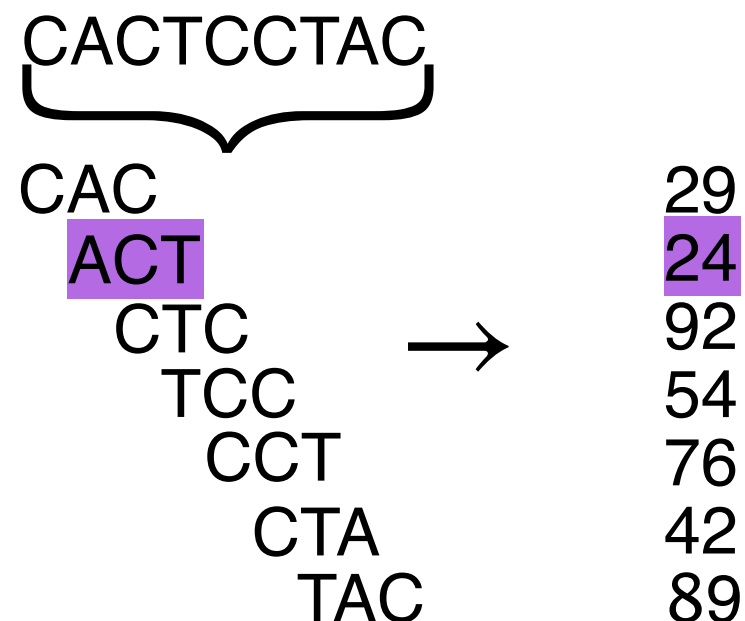
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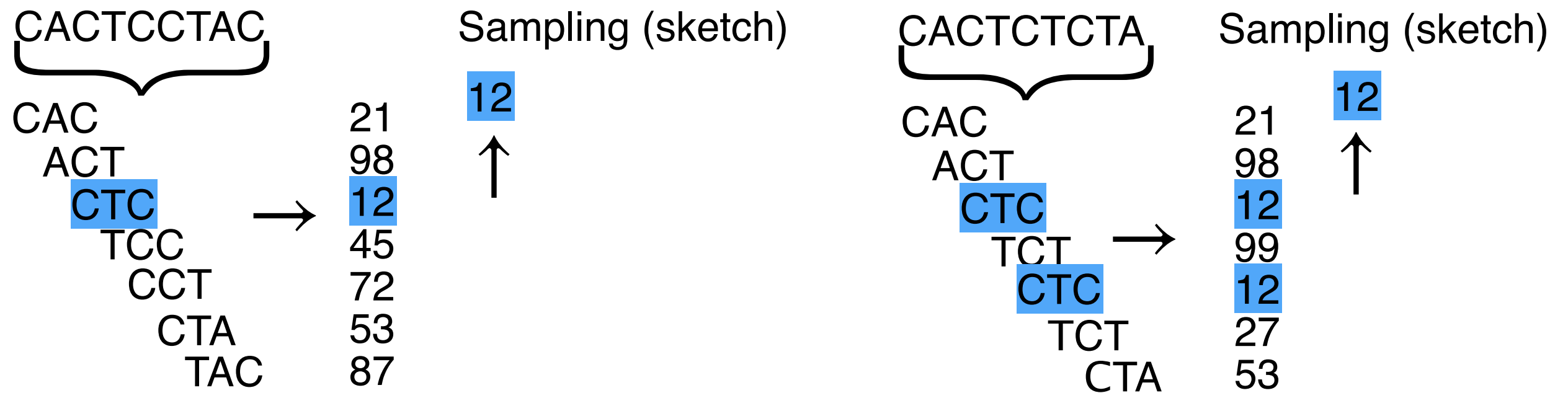


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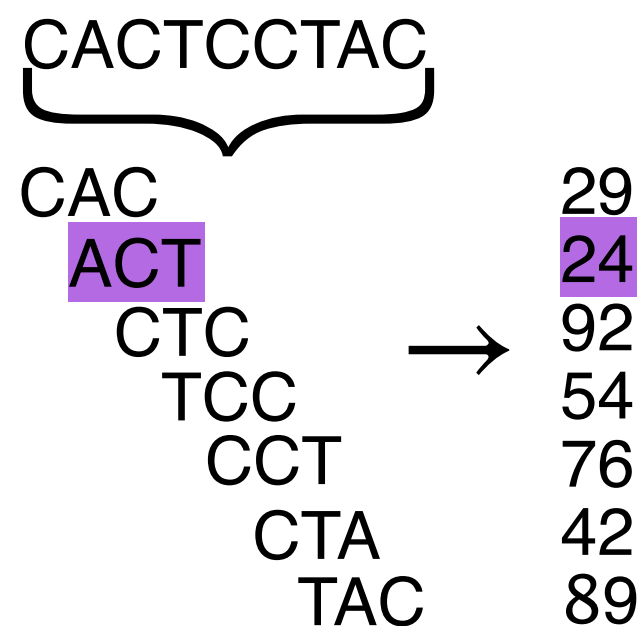
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MinHash sketching: Comparing sketches

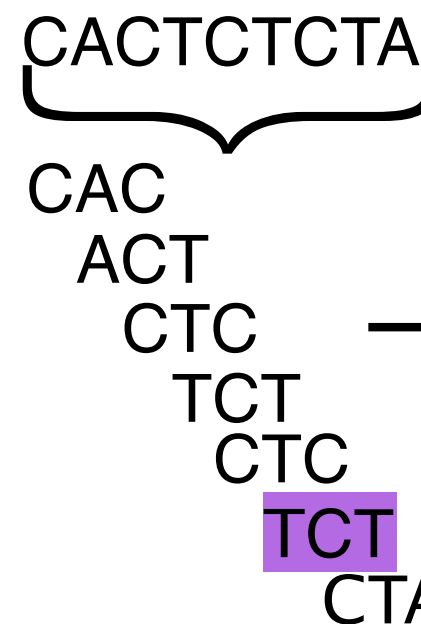


MinHash sketching: Comparing sketches



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Sampling (sketch)

12, 9, 19, 13, 6

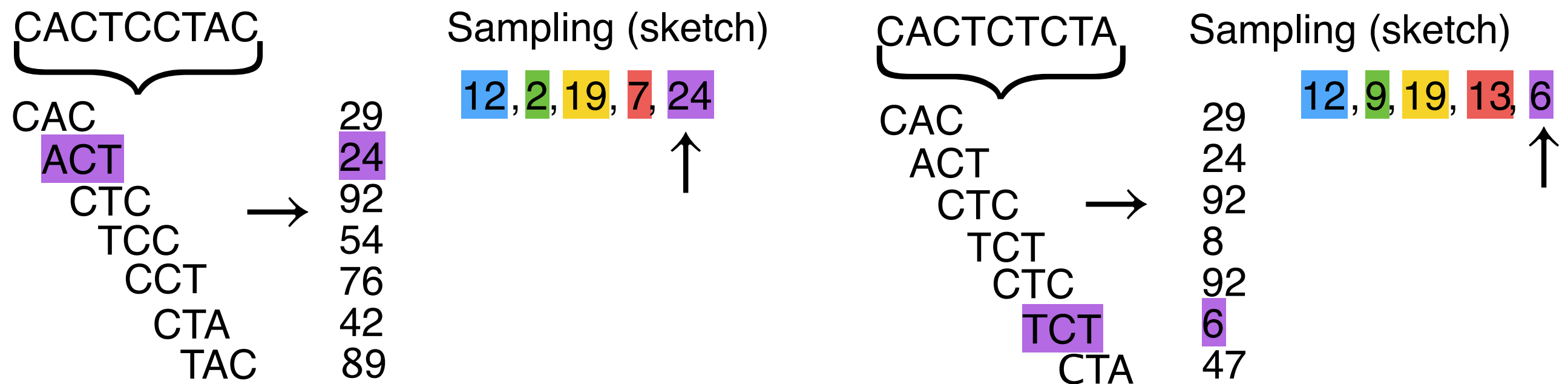


Sketch comparison:

12, 2, 19, 7, 24
| × | × ×
12, 9, 19, 13, 6

$$= \frac{2}{5}$$

MinHash sketching: Comparing sketches



MinHash fundamental property: The fraction of matching MinHash values is an unbiased estimate of the Jaccard index

What have we achieved?

Comparing long sequences with many k-mers:

Computationally intense!

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Estimates Jaccard index

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	×		×	×
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2. How do we implement MinHash in practice?
3. Unbiased estimate of Jaccard index?

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Hash function

Example: The item is a k -mer ACTTG

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2. Apply a function $h_p(x) = (p \cdot x) \bmod N$

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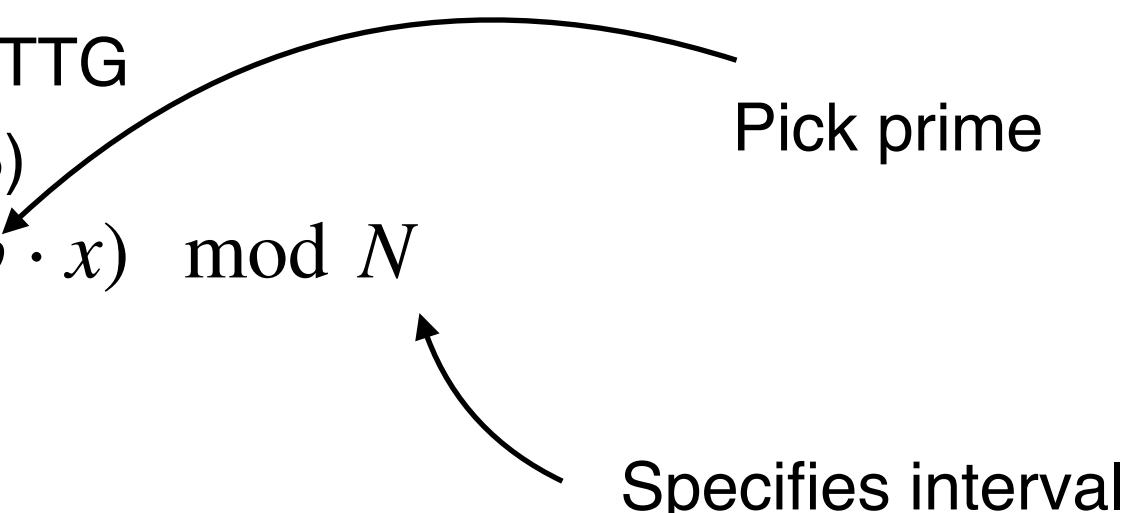
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Specifies interval

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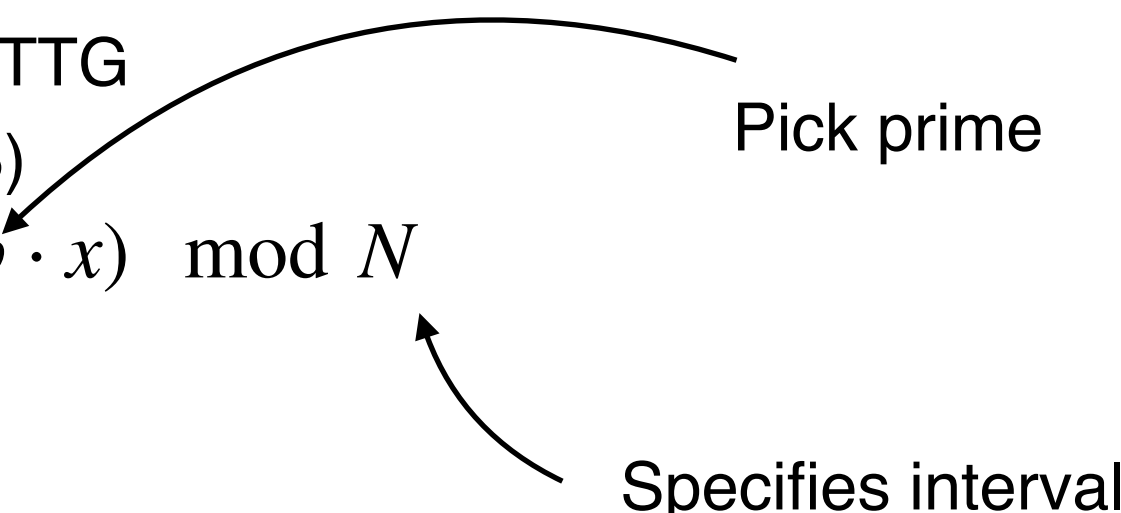
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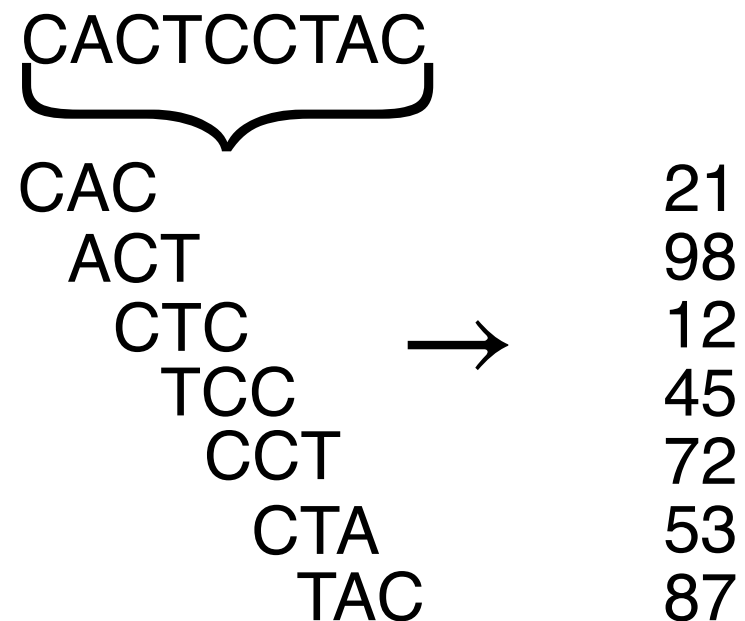
Specifies interval

Pick primes: $p_1, \dots, p_m$ for m "mappings"

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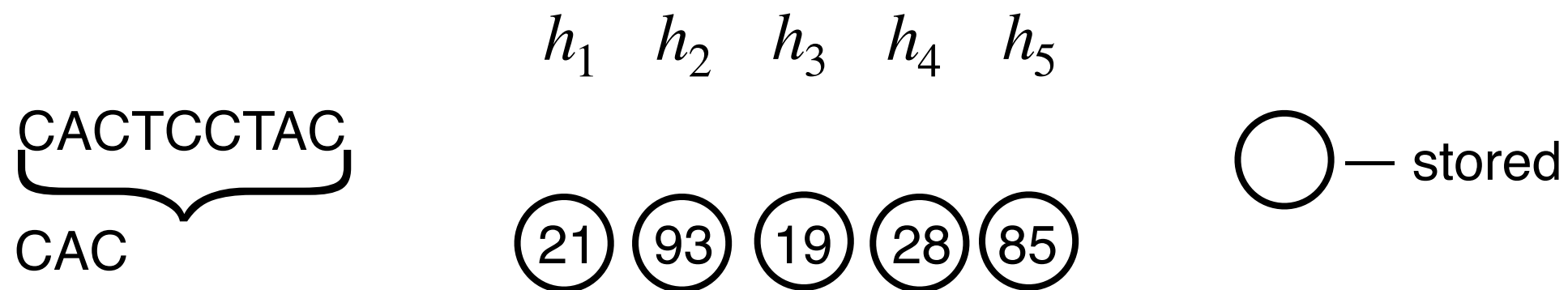
Streaming one k -mer at a time



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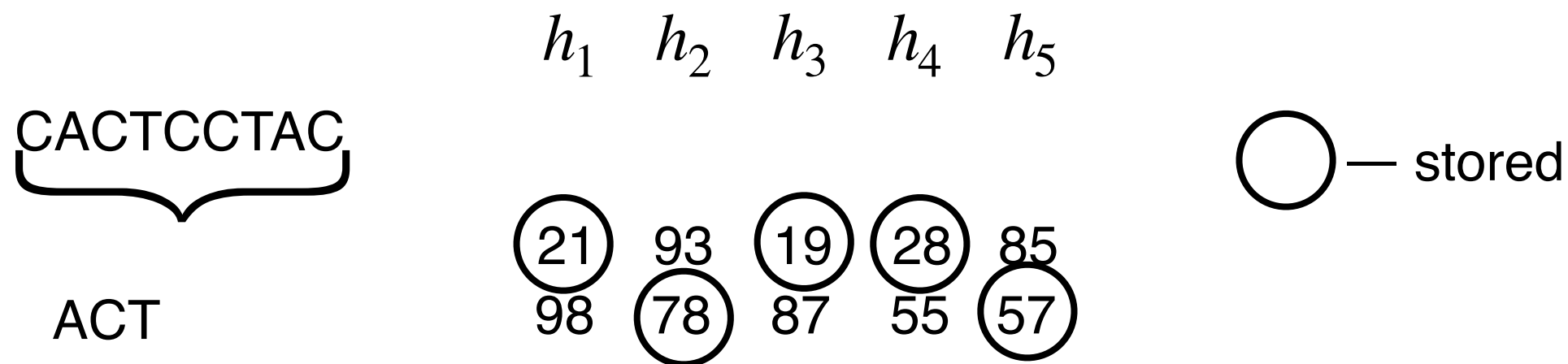
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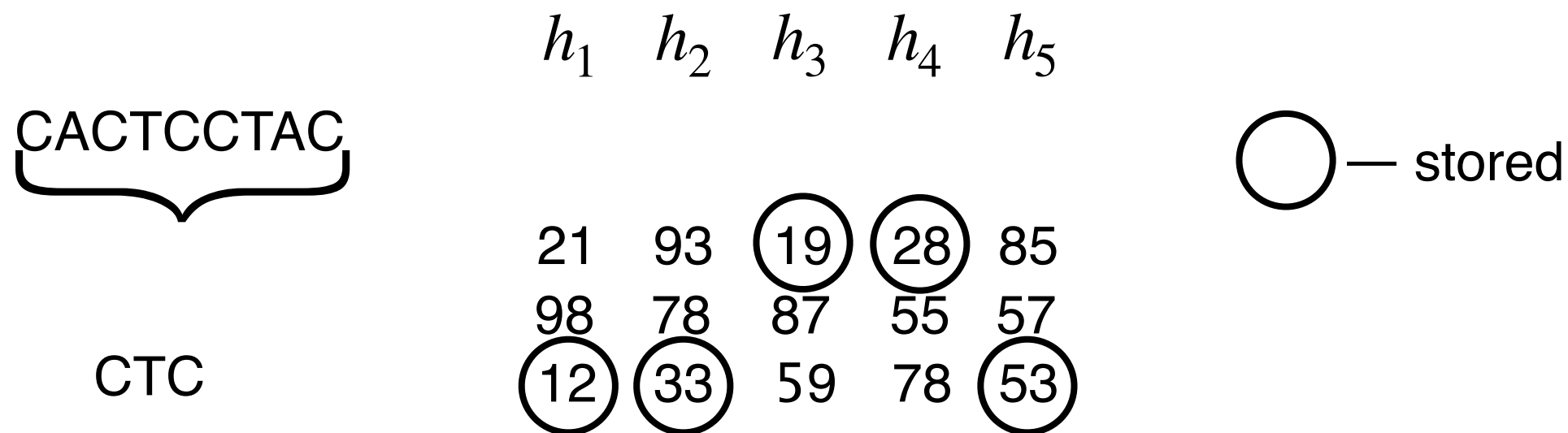
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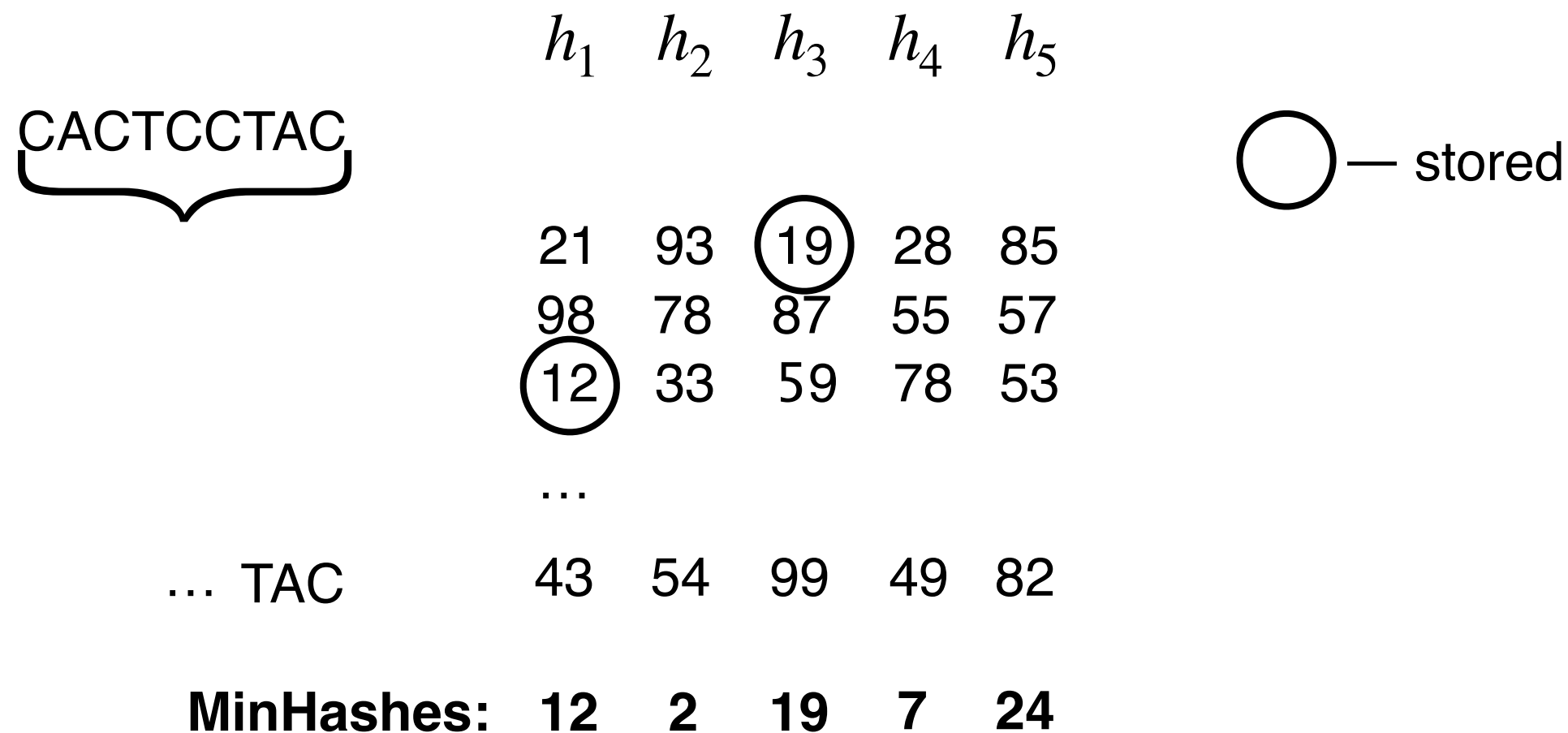
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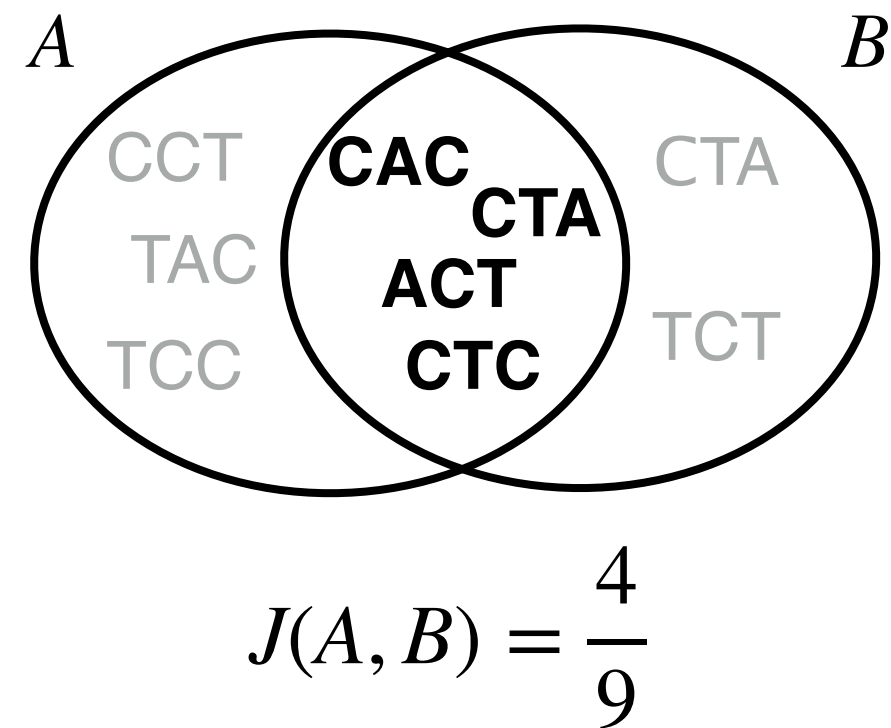
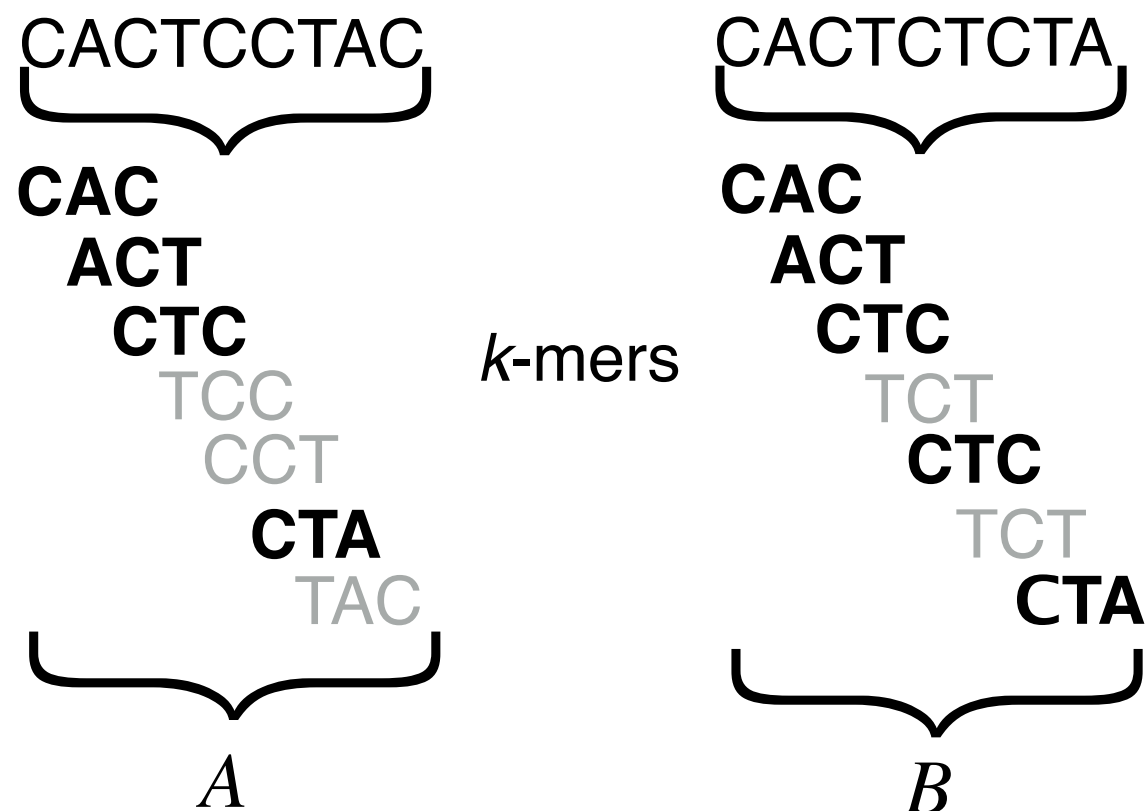
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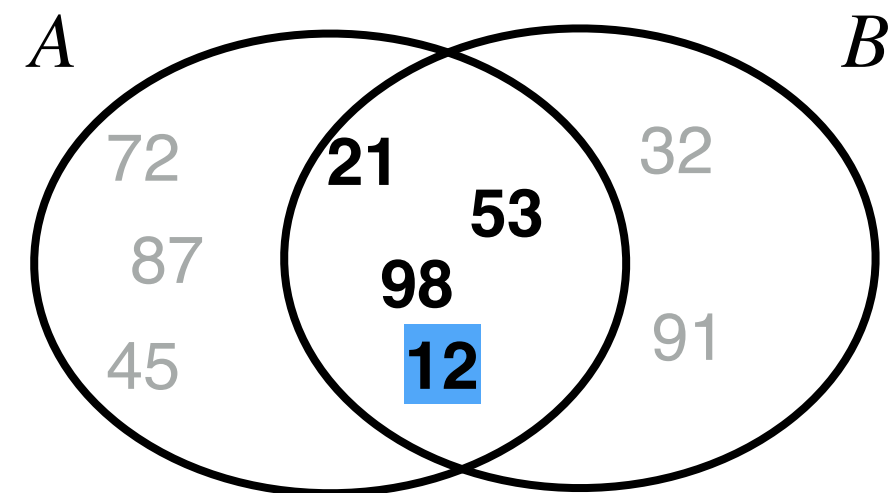
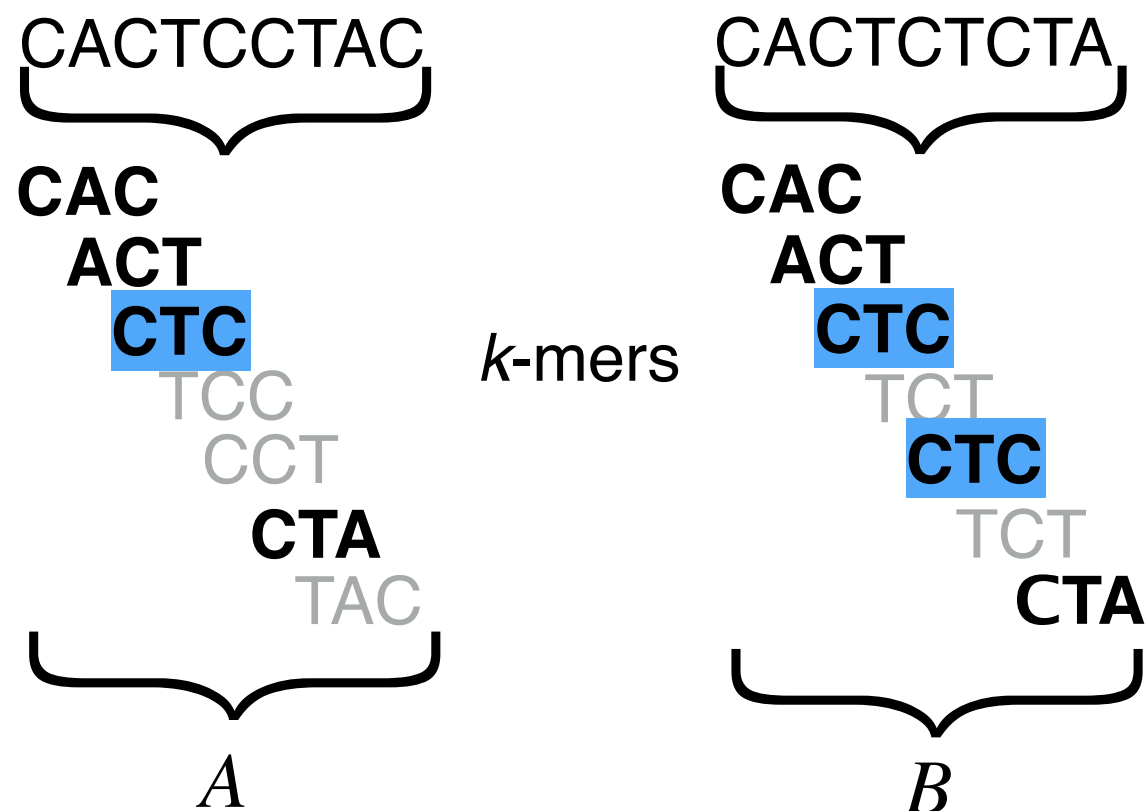
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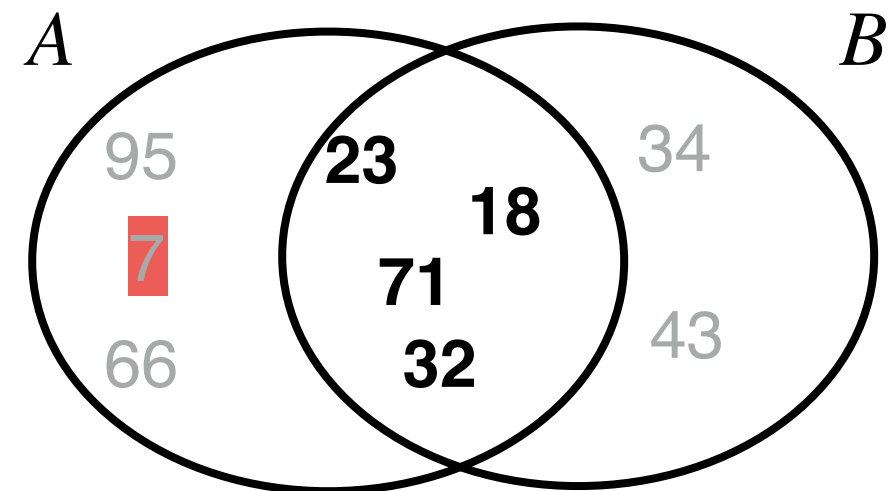
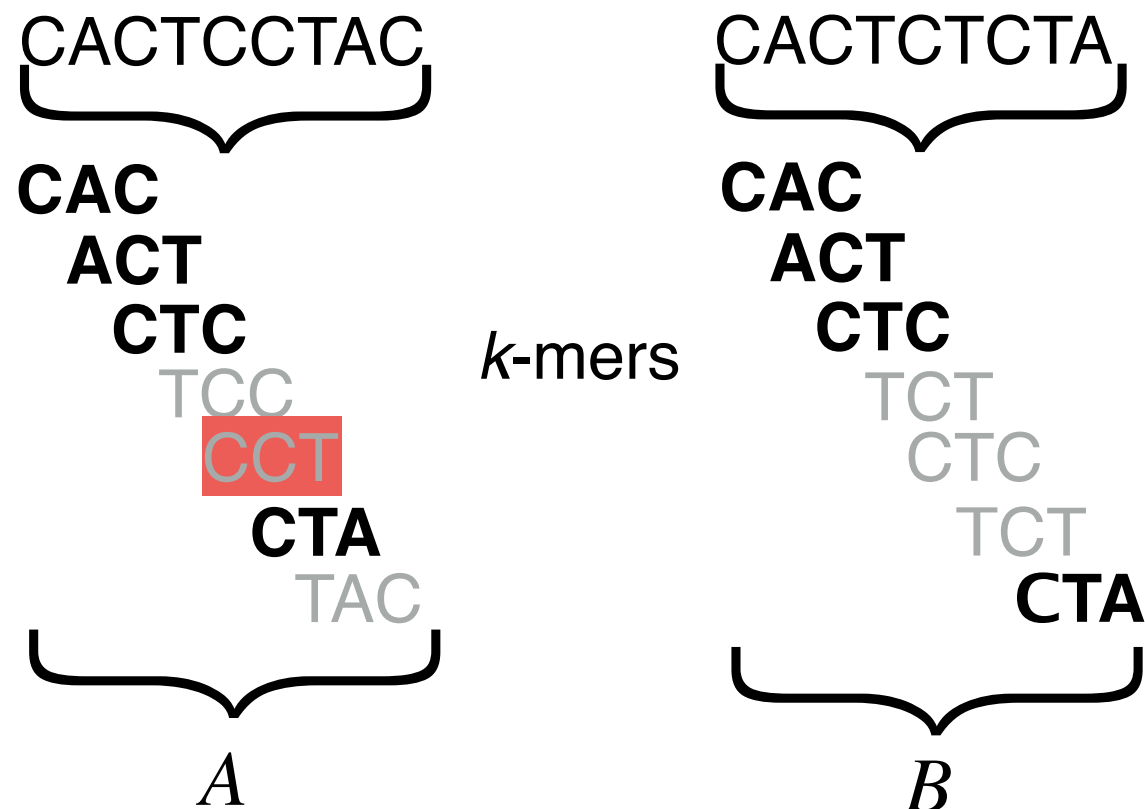


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$$\hat{J}(A, B) = \frac{1}{m} \sum_{i=1}^m X_i$$

MinHash in practice

$m : 1,000 - 100,000, \quad k : 15 - 60, \quad N = 2^{32}$

MinHash in practice

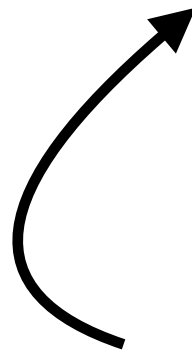
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:
:
n TGTTTTTACGACTCTGGTACCTAGACTCGACGATCGTACCCTGCTGCTGTCAGATCGAGCCTCACTA....

New sample

TTACGACTCTGGTACCTACGTGCAGTATCTATGCACACAGAAAAACAGTCTACAGCGTATGCCGAGCATCGTA....



Similar to???

Index the database

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ID	h_1	h_2	h_3	...	h_m
0	9,201	3,454	19	...	9,344
1	1,204	1,455	65,334	...	43,563
2	342	1,455	8,891	...	97
3	80,009	13,677	243	...	876
...	...	...	...	...	...
n	9,201	344	8,273	...	3,627

Search new sequence

New sample

TTACGACTCTGGTACCTACGTGCAGTATCTATGCACACAGAAAAACAGTCTACAGCGTATGCGAGCATCGTA....

h_1	h_2	h_3	...	h_m
23,301	13,677	243	...	745

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$n = 1,000,000$
 $m = 1,000$
 Integer: 4 byte
 4 Gigabyte index

Bacterial seq ~
 4,000,000 letters
 4 Terabyte index

Inverted index (Hash table)

Instead of storing: "*Bacteria ID* → *Minhashes it contains*",
we flip it to: "*MinHash* → *list of bacteria containing it*"

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2	342	1,455	...	97
3	80,009	13,677	...	876
...	...	...	...	...
n	9,201	344	...	3,627

To

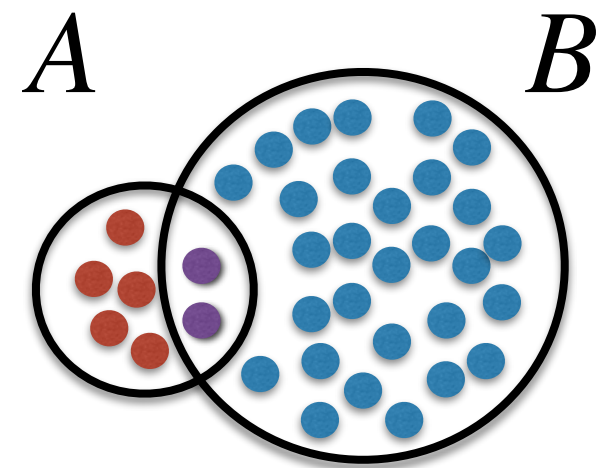
Hash value	Bacterial IDs
1,204	1 244,659
97	74 823 48
1,455	1 2 52 1,567
4,911	5 281
...	

Is MinHash good for your application?

- MinHash does not handle multisets
 - Repetitive k-mers: ACG**ACG**
 - Hash collisions $h(\text{ACG}) = h(\text{CGA})$

Is MinHash good for your application?

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 - Hash collisions $h(\text{ACG}) = h(\text{CGA})$
- *Need large sample size (m) when intersection is small*
 - *Dissimilar*
 - *One set much smaller than the other*



Limitations with MinHash and some variations

1. Computation time of m hash functions

Solution: ***bottom MinHash***:

- Use one hash function h_1
- Store the m smallest hash values

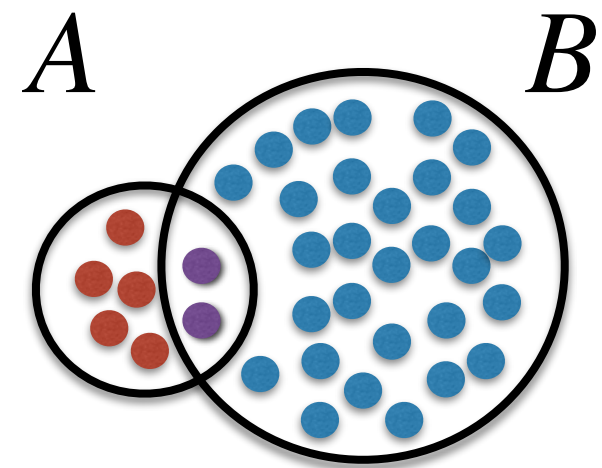
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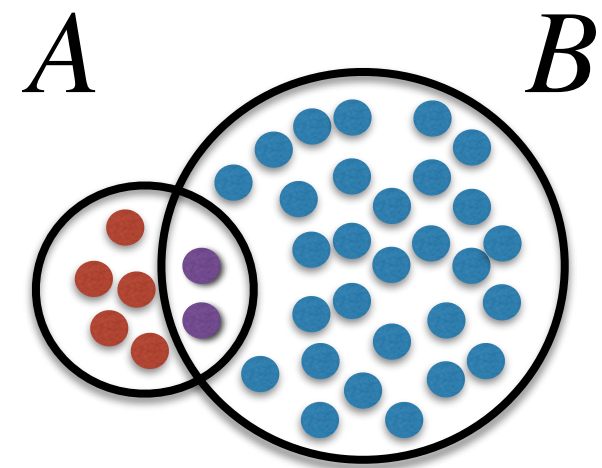
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2. Different sized sequences

Solution: **Scaled *MinHash*** (*FracMinHash*):

- Use one hash function h_1
- Ex. Store $h(v)$ if in $[0, X]$, discard $[X+1, N]$
- Unbounded sketch size



Summary

Problem: Need to compare texts

- Retrieval of similar biological sequences (, documents, webpages)

How to measure similarity?

- Hamming/edit distance: *No unordered local similarities!*
- Jaccard Index based on *k*-mers: *Finds local similarities! But computationally intense!*

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Sampling of large databases

- *MinHash Sketching*
 - Streaming (on-the-fly)
 - Unbiased estimate of Jaccard Index

Real life scenario

- A large database is *sketched* and stored in memory as index
- New data is obtained, sketched and queried
- Top matches are investigated further

Module assignment

- **Task 1:** Sketch genomes and compute $d = 1 - J$ between all pairs.
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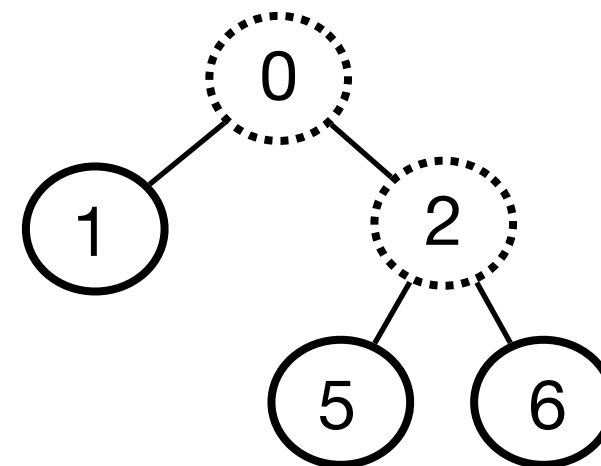
Accession:
starts with '>'

Sequence

```
>1 parent lineage: [0]
GGAAACGCTATTATGCGTTTAGTCCTTACCGCGTCCGGCTACATGTTACCTTTTGTTCATCCTTTTCGGCCC
AGGGTCCGTTGCCGTACGTTTCAGCAGTGGTGAATCGTCCGACCCCAGGATTGCTACATCATGCTCCCAT
ATGCGACGTACAAAACGAGACTCCCGGCCTCATACTTCTTTATTGAAGACGGTGCCCTAT
>5 parent lineage: [2, 0]
GGAAACGCTAATATGCGTTTAGTCCTTACCGCGTCCGGGTACATGTTACCTTTTGTTCATCCTTTTCGGCCC
AGGGTCCGTTGCCCTACGTTACAGCAGTGGTGAATCGTCCGACCCCAGGATTGCTACATGATGGTCCCAT
ATGCGACGTACAAAACGAGAGTCCCGGCCACAAAGTTCTATATTGAAGACGGTGCCCTAT
>6 parent lineage: [2, 0]
GGAATCGCTATTATGCGTTTAGTCCTTAGCGCGTCCGGGTACATGTTACCTTTTGTTCATCCTTTTCGGCCC
AGGGTCCGTTGCCCTACGTTACAGCAGTGGTGAATCGTCCGACCCCAGGATTGCTACATGATGGTCCCAT
ATGGGACGTACAAAAGGAGACTCCCGGCCACAACTTCTTTATTGAAGACGGTGCCCTAT
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Shown in simulated data
for debugging

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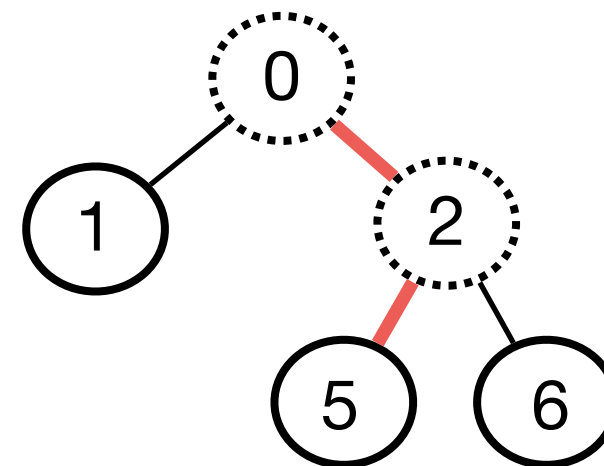
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>1 parent lineage: [0]
GGAACGCTATTATGCGTTTAGTCCTTACCGCGTCCGGCTACATGTTACCTTTTGTTCATCCTTTTCGGCCC
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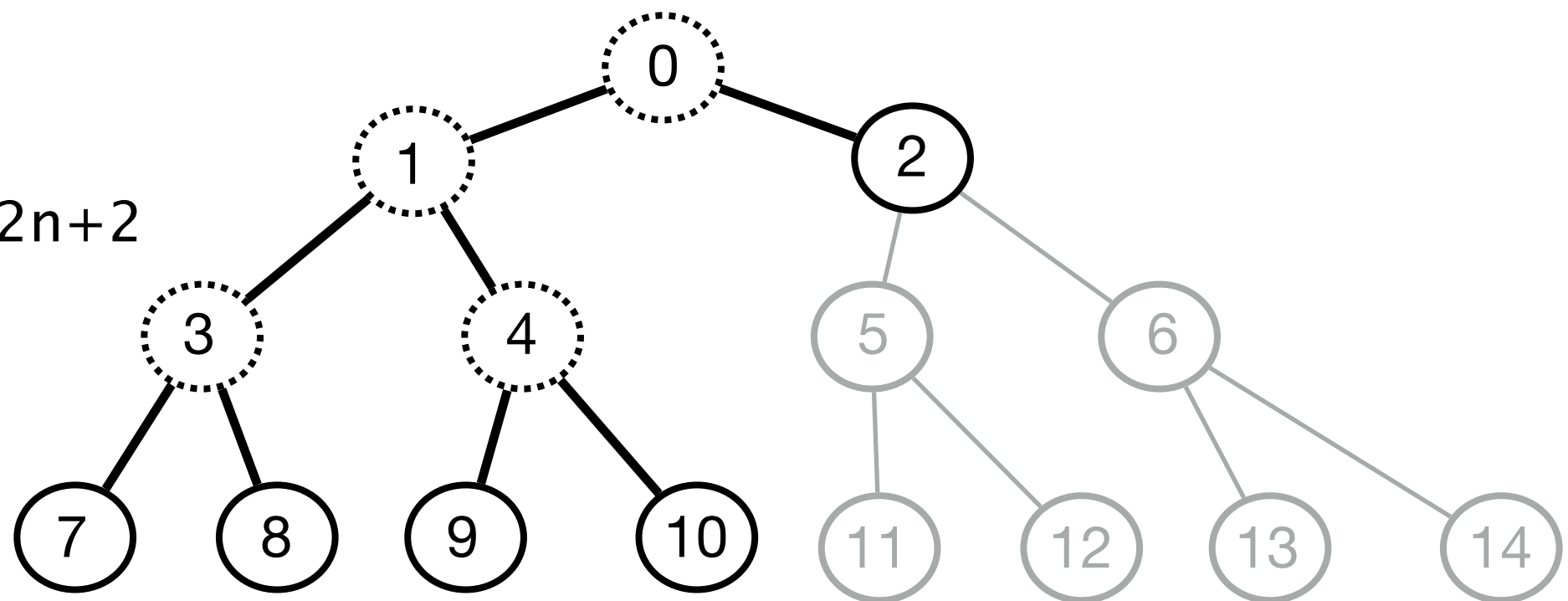
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GGAAACGCTATTATGCGTTTAGTCCTTACCGCGTCCGGCTACATGTTACCTTTTGTTCATCCTTTTCGGCCC
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Parent: n

Child labels: $2n+1, 2n+2$



Deadline: September 23rd, 23:59

Appendix

Some notes

- MinHash does not handle multisets
 - Repeated items can occur due to
 - Repetitive k-mers
 - Hash collisions
- Why Jaccard index in previous example?
 - *Set lower cut-off in matching identity (depends on k , m).*
 - *Estimate number of random hits (depends on k , n)*
 - *One can compute Average Nucleotide Identity (0-100%) from J , k .*

Space and time complexity?

- **Indexing**
 - $O(nm)$ space complexity.
 - $O(nmK)$ time complexity, for K k -mers
- **Querying**
 - $O(nm)$ time complexity.

Inverted index (Hash table)

- **Indexing**
 - $O(nm)$ space complexity.
 - $O(nmK)$ time complexity, for K k -mers
- **Querying**
 - $O(mK)$ - compute Minhash sketch
 - $O(m)$ - look up m hash values in index
 - $O(??)$ - merge the candidate bacteria ID lists
 - Depends on how similar Bacterias are (worst case $O(nm)$)
 - $O(Lm)$ if L average length of lists, $L \ll n$.

MinHash¹

Universe of items $\mathcal{U}$

Ex. $\mathcal{U} = \{A, B, C, D, E\}$

Permutation $\pi : \mathcal{U} \rightarrow \mathcal{U}$

Ex. Shuffling of universe:
 $\pi(\mathcal{U}) = [C, A, E, D, B]$

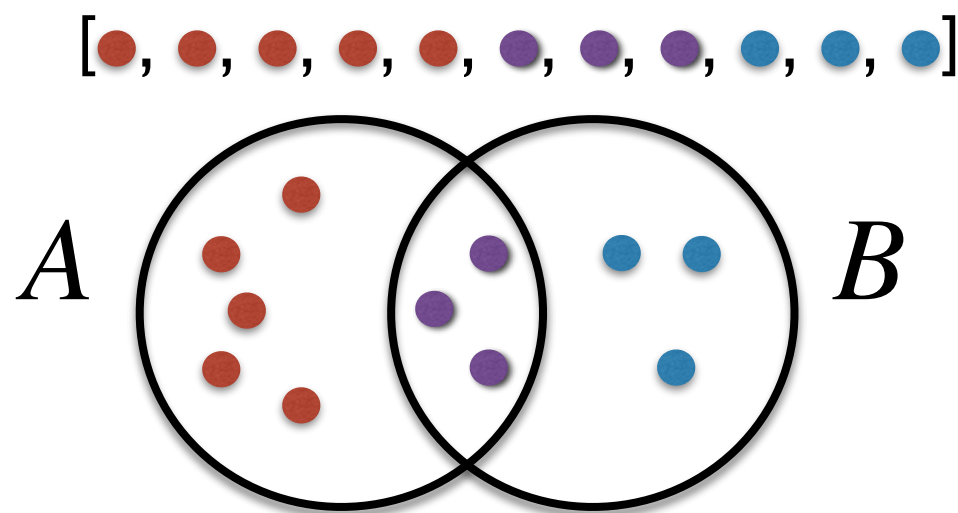
MinHash: $M_\pi(S) = \min_{x \in S} \pi(x)$, $S \subseteq \mathcal{U}$.

Ex. Item with smallest index in π .

MinHash fundamental property:

$$P(M_\pi(S) = M_\pi(T)) = \frac{|S \cap T|}{|S \cup T|}$$

Ex. $S = \{A, C, D\}$, $T = \{C, D, E\}$
Jaccard = 1/2



MinHash theory: Probability, one permutation

Estimator construction: m independent permutations

Implementation: m hash functions

[1] A. Z. Broder, 1997 "On the resemblance and containment of documents"

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- [1] A. Z. Broder, "On the resemblance and containment of documents," *Proceedings. Compression and Complexity of SEQUENCES 1997 (Cat. No.97TB100171)*, Salerno, Italy, 1997, pp. 21-29, doi: 10.1109/SEQUEN.1997.666900.
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